

Cybersecurity Toolkit

Hands-on Tutorial for the Arch Linux VM

BlackArch tools on Arch Linux • Hyprland desktop
Login: arch / arch

0. Read This First — Legal & Ethics

These tools are for **authorized testing and learning only**. Only scan, probe, or attack systems you **own** or have **explicit written permission** to test. Unauthorized access or scanning is illegal in most countries (e.g. the U.S. CFAA, the UK Computer Misuse Act).

Build a safe lab instead of touching real targets:

- Practice VMs: Metasploitable2/3, OWASP Juice Shop, DVWA, VulnHub images.
- Online labs: TryHackMe, Hack The Box, PortSwigger Web Security Academy.
- Keep targets on an isolated host-only network.

1. How This VM Is Set Up

The BlackArch repository (~5,000 security packages) and multilib are enabled on top of Arch. A broad toolkit is already installed. To add more tools later:

```
# search the security repo
pacman -Ss <keyword>

# install a tool
sudo pacman -S <tool>

# install a whole BlackArch category (large!)
sudo pacman -S blackarch-scanner # or -webapp, -cracker, -wireless ...
```

Wordlists live in `/usr/share/seclists` and `/usr/share/wordlists` (rockyou.txt is the classic).

2. Reconnaissance & Information Gathering

Map what exists before touching it: hosts, ports, DNS, subdomains, services.

nmap

The core network/port scanner.

```
nmap -sV -sC -oN scan.txt 192.168.56.0/24 # service+script scan a subnet
nmap -p- --min-rate 5000 10.0.0.5 # all 65535 ports, fast
```

masscan / rustscan

Internet-speed port sweeps, then hand off to nmap.

```
rustscan -a 10.0.0.5 -- -sV
masscan 10.0.0.0/24 -p1-65535 --rate 10000
```

dnsrecon / fierce / amass

Enumerate DNS records and subdomains.

```
dnsrecon -d example.com
amass enum -d example.com
```

whatweb / wafw00f

Fingerprint web tech and detect WAFs.

```
whatweb https://example.com
wafw00f https://example.com
```

enum4linux / netexec

Enumerate SMB/Windows shares, users, policies.

```
enum4linux -a 10.0.0.10  
netexec smb 10.0.0.0/24 -u user -p pass --shares
```

3. Web Application Testing

Find and exploit common web flaws: injection, XSS, hidden content.

sqlmap

Automated SQL injection detection & exploitation.

```
sqlmap -u 'http://site/item?id=1' --batch --dbs  
sqlmap -u '...' --batch --dump -D appdb -T users
```

gobuster / ffuf / feroxbuster

Brute-force directories, files, vhosts.

```
gobuster dir -u http://site -w /usr/share/seclists/Discovery/Web-Content/common.txt  
ffuf -u http://site/FUZZ -w wordlist.txt
```

nikto

Quick web server vulnerability scan.

```
nikto -h http://site
```

wpscan

WordPress-specific scanner.

```
wpscan --url http://site --enumerate u,vp
```

4. Exploitation

Turn a known weakness into access — in your lab only.

Metasploit Framework

The standard exploitation framework.

```
msfconsole  
search type:exploit smb  
use exploit/...  
set RHOSTS 10.0.0.5  
run
```

searchsploit (exploitdb)

Offline Exploit-DB search.

```
searchsploit apache 2.4  
searchsploit -m 50383 # copy an exploit locally
```

impacket / netexec

Windows/AD protocol toolkit & lateral movement.

```
impacket-secretsdump domain/user:pass@10.0.0.10  
netexec smb 10.0.0.10 -u u -p p -x whoami
```

evil-winrm

Interactive shell over WinRM.

```
evil-winrm -i 10.0.0.10 -u admin -p pass
```

5. Password Attacks

Crack hashes offline or brute services online (lab only).

hashcat

GPU-accelerated hash cracking.

```
hashcat -m 0 hashes.txt /usr/share/seclists/Passwords/rockyou.txt # MD5  
hashcat -m 1000 ntlm.txt rockyou.txt # NTLM
```

hydra / medusa / ncrack

Online brute force of network logins.

```
hydra -l admin -P rockyou.txt ssh://10.0.0.5  
hydra -L users.txt -P pass.txt ftp://10.0.0.5
```

hashid / hash-identifier

Identify an unknown hash type.

```
hashid '5f4dcc3b5aa765d61d8327deb882cf99'
```

crunch / cewl

Generate custom wordlists.

```
crunch 8 8 -o list.txt  
cewl http://site -w site-words.txt
```

6. Wireless

Audit Wi-Fi (needs a USB adapter passed into the VM that supports monitor mode).

aircrack-ng suite

Capture handshakes and crack WPA.

```
airmon-ng start wlan0  
airodump-ng wlan0mon  
aircrack-ng -w rockyou.txt capture.cap
```

wifite

Automated wireless auditing wrapper.

```
wifite
```

bettercap

Swiss-army network/MITM/wireless tool.

```
bettercap -iface wlan0
```

7. Sniffing & MITM

Inspect and manipulate traffic on a network you control.

Wireshark / tcpdump

Capture and analyze packets.

```
tcpdump -i eth0 -w cap.pcap  
wireshark cap.pcap
```

ettercap / bettercap

ARP spoofing and man-in-the-middle.

```
ettercap -T -M arp /10.0.0.5// /10.0.0.1//
```

mitmproxy

Intercept and edit HTTP(S).

```
mitmproxy
```

8. Reverse Engineering & Forensics

Analyze binaries, carve files, inspect memory and disks.

radare2 / rizin

Disassemble and debug binaries.

```
r2 -A ./binary  
# inside: aaa; afl; pdf @main
```

binwalk / foremost

Analyze firmware, carve embedded files.

```
binwalk -e firmware.bin  
foremost -i disk.img -o out/
```

volatility3

Memory-image forensics.

```
vol -f mem.raw windows.pslist
```

steghide / stegseek

Hide/recover data in images (stego).

```
steghide extract -sf image.jpg  
stegseek image.jpg rockyou.txt
```

exiftool

Read/strip file metadata.

```
exiftool photo.jpg
```

9. Anonymity & Utilities

Route tools through proxies/Tor and pivot.

proxychains-ng + tor

Route any tool through Tor/proxies.

```
sudo systemctl start tor  
proxychains nmap -sT 10.0.0.5
```

socat / netcat

Swiss-army networking, shells, port relays.

```
nc -lvnp 4444 # listener  
socat TCP-LISTEN:4444,fork TCP:10.0.0.5:80
```

10. A Typical Engagement Flow

1. **Recon** — discover hosts/ports/services (nmap, masscan).
2. **Enumerate** — dig into each service (enum4linux, whatweb, gobuster).
3. **Find weaknesses** — version/CVE lookup (searchsploit, nuclei, nikto).
4. **Exploit** — gain access in your lab (metasploit, sqlmap).
5. **Post-exploit** — loot/pivot (impacket, netexec, mimikatz-style).
6. **Crack** — offline hash cracking (hashcat, john).
7. **Report** — document findings and remediation.

Learn More

- PortSwigger Web Security Academy (free)
- TryHackMe / Hack The Box
- OWASP Testing Guide
- BlackArch tool list: blackarch.org/tools.html